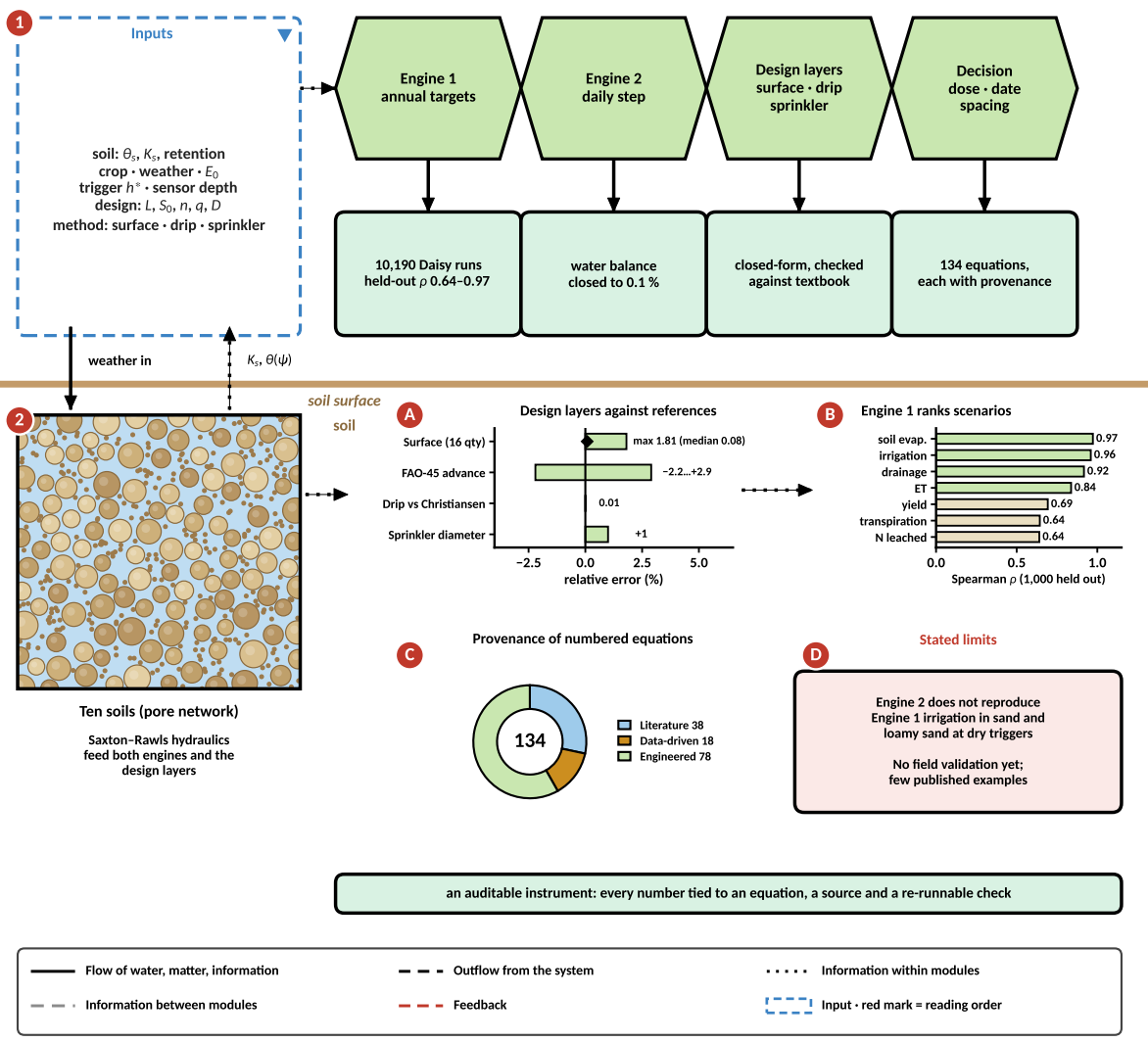

1
Inputs
soil: θ_s, K_s, retention
crop · weather · E_0
trigger h^* · sensor depth
design: L, S_0, n, q, D
method: surface · drip · sprinkler
Engine 1
annual targets
Engine 2
daily step
Design layers
surface · drip
sprinkler
Decision
dose · date
spacing
10,190 Daisy runs
held-out ρ 0.64–0.97
water balance
closed to 0.1 %
closed-form, checked
against textbook
134 equations,
each with provenance
weather in
K_s, $\theta(\psi)$
2
soil surface
soil
Ten soils (pore network)
Saxton–Rawls hydraulics
feed both engines and the
design layers
A
Design layers against references
Surface (16 qty)
max 1.81 (median 0.08)
FAO-45 advance
−2.2...+2.9
Drip vs Christiansen
0.01
Sprinkler diameter
+1
−2.5
0.0
2.5
5.0
relative error (%)
B
Engine 1 ranks scenarios
soil evap.
0.97
irrigation
0.96
drainage
0.92
ET
0.84
yield
0.69
transpiration
0.64
N leached
0.64
0.0
0.5
1.0
Spearman ρ (1,000 held out)
C
Provenance of numbered equations
134
Literature 38
Data-driven 18
Engineered 78
D
Stated limits
Engine 2 does not reproduce
Engine 1 irrigation in sand and
loamy sand at dry triggers
No field validation yet;
few published examples
an auditable instrument: every number tied to an equation, a source and a re-runnable check
Flow of water, matter, information
Outflow from the system
Information within modules
Information between modules
Feedback
Input · red mark = reading order